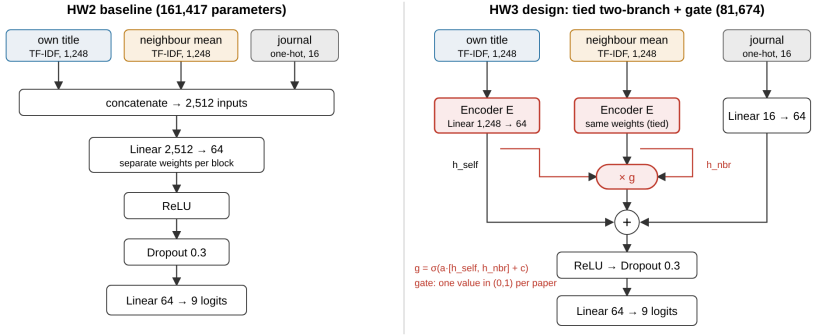
HW3 – Break the Network: a weight-tied two-branch encoder with a learned neighbour gate

Yingrong Mao · DS7400 Deep Learning · Repository: github.com/MaoYingrong/DS7400_deep_learning

1. Baseline (HW2)

Task and data. Node classification on a 2,000-paper sample of the APS citation network: predict a paper's PhySH discipline (9 classes). Inputs are the title TF-IDF (1,248 terms), the mean title TF-IDF of its citation neighbours (1,248) and a journal one-hot (16), concatenated. **Model.** Linear(2,512→64) → ReLU → Dropout 0.3 → Linear(64→9), 161,417 parameters. **Training.** Cross-entropy, Adam (lr 0.001, weight decay 0.001), batch 64, 40 epochs, weights from the epoch with the lowest validation loss; stratified 1,400/300/300 split. **Metric.** Test accuracy and macro-F1. The baseline reaches 77.2% test accuracy.



2. Limitation and proposed design

Limitation. The baseline overfits (92.7% train vs 74.7% validation accuracy), and 99.6% of its parameters are in the first layer. That layer learns two unrelated weight matrices for the paper's own title and its neighbours' titles although both live in the same vocabulary, and it cannot choose how much to trust neighbours for a given paper.

Change. (1) Both text inputs pass through one shared encoder E (weight tying), halving the parameters to 81,674. (2) The neighbour branch is added as a residual-style correction scaled by a learned per-paper gate: $h = \mathrm{ReLU}(E(x_{self}) + g \cdot E(x_{nbr}) + W_j x_{journal})$, with $g = \sigma(a \cdot [h_{self}, h_{nbr}] + c)$. With untied encoders and g = 1 this is exactly the baseline, so each ingredient can be switched off separately.

> **Original hypothesis (before running).** Tying will reduce overfitting and raise test accuracy, because a word carries the same field information in a paper's own title and in its neighbours' titles. The gate will learn to close for papers whose neighbours are uninformative (few or mixed-field neighbours). **Possible failures:** a mean over many titles is distributed differently from a single title, so shared weights may fit neither well; and the sigmoid gate may saturate and become a constant.

3. Experiments

Features, split, optimiser, learning rate, batch size, epochs and model selection are identical for all variants; only the model changes. Each variant is trained with 10 seeds (initialisation and batch order). Besides my design I ran four ablations (remove the gate; untie the weights; replace the per-paper gate by one global number; remove the neighbour branch) and an 80-epoch run to check the training budget.

Variant	Params	Train acc %	Val acc %	Test acc %	Macro-F1	Train−val gap	Test diff vs baseline
HW2 baseline (untied, fixed add)	161,417	92.7	74.7 ± 0.6	77.2 ± 0.3	0.613	18.0	–
Tied + per-paper gate (my design)	**81,674**	**89.9**	**76.4 ± 0.5**	**76.3 ± 0.7**	**0.610**	**13.5**	**−0.9 ± 0.3**
Tied, fixed add (no gate)	81,545	89.6	76.2 ± 0.5	76.3 ± 0.7	0.606	13.4	−0.9 ± 0.2
Untied + per-paper gate	161,610	92.0	74.2 ± 0.6	76.6 ± 0.6	0.606	17.7	−0.6 ± 0.2
Tied + one global gate	81,546	90.5	75.1 ± 0.8	75.4 ± 0.9	0.594	15.4	−1.8 ± 0.3
Tied, neighbour branch removed	81,545	90.4	73.7 ± 0.5	72.1 ± 0.6	0.556	16.7	−5.1 ± 0.2

Mean ± standard deviation over 10 seeds; last column is the paired mean difference ± standard error. 80-epoch check: baseline 77.2, tied + gate 76.4, tied 76.5.

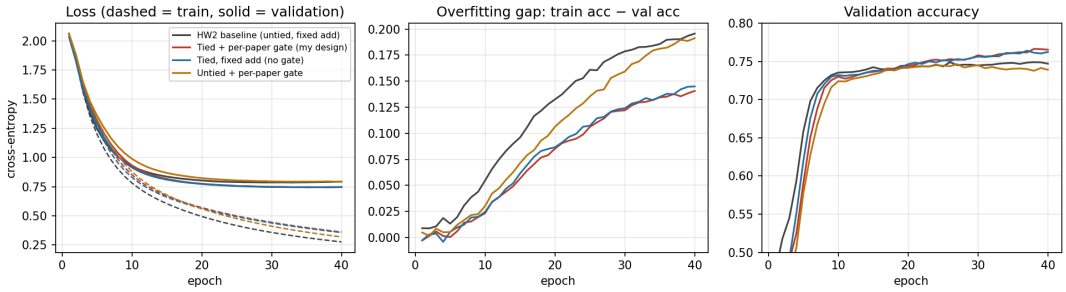


Figure 1. Mean over 10 seeds. Tied models (red, blue) have a smaller overfitting gap and lower validation loss; adding the gate changes almost nothing.

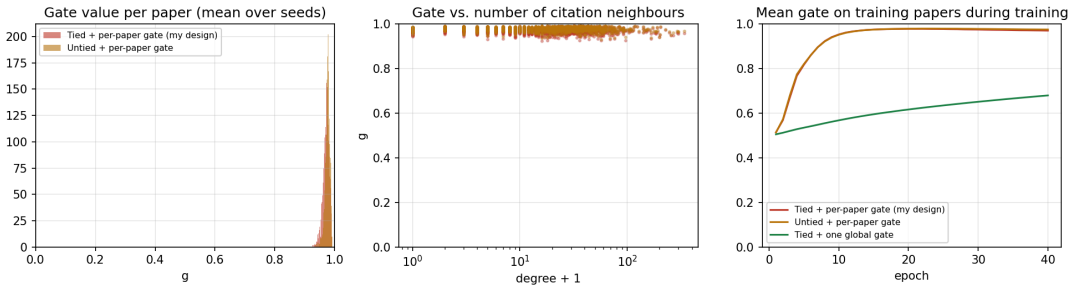


Figure 2. The gate opens to about 0.97 for every paper regardless of how many neighbours it has. The single global gate moves much more slowly.

4. Observations (after running) versus the hypothesis

> **Summary.** Half of the hypothesis held in a weak form and half failed. Tying halved the parameters and reduced the overfitting gap, but did not improve test accuracy. The gate saturated open and did nothing.

- **Overfitting: partly as expected.** The train−validation gap fell from 18.0 to 13.5 points and validation accuracy rose 1.7 points. Most of the smaller gap comes from lower training accuracy (92.7 → 89.9), i.e. less memorisation capacity.
- **Test accuracy: not as expected.** It fell 0.9 points, about 3 of 300 papers. Validation and test move in opposite directions, and seeds do not capture the noise of the single fixed split, so I conclude only that accuracy is about the same with half the parameters.
- **Gate: failed, in the predicted way.** g rose from 0.5 to about 0.97 within roughly 10 epochs for all papers (standard deviation across papers 0.011; correlation with log degree −0.02; 0.973 on correct vs 0.966 on wrong test papers). Tied + gate and tied without gate have the same test accuracy (76.3).
- **Why it saturated.** Neighbours are the most useful input (removing them costs 5.1 points), so the gradient always pushes the gate open, and a sigmoid can only shrink the branch, never amplify it. The gate had no useful decision to make.
- **Unexpected: the global gate was the worst neighbour-using variant (−1.8).** Its one parameter starts at 0.5 and Adam changes it by at most about the learning rate per step, so after 880 steps it had only reached 0.68. The per-paper gate has 129 parameters acting on features and opened quickly. This is an optimisation artefact of my initialisation, not evidence about gating itself.
- **Convergence and cost.** All variants trained stably. Tied models reach their best epoch later (36.8 vs 31.5), but 80 epochs gave the same ranking. Parameter memory is halved; time per run rose from 1.07 s to 1.35 s because of the second encoder pass.
- **Alternative explanation to check.** Papers with more than 30 neighbours scored higher without the neighbour branch (80.9% vs 78.4%, n = 58), which is where a working gate should close. That is too few papers to trust, but it suggests the idea is not wrong, only that my gate could not find it.

5. What I learned and what I would do next

A learned gate only helps if the loss gives it a reason to differ across examples; here one global answer ("use the neighbours") was good enough, so the extra mechanism collapsed to a constant. Weight sharing is a real regulariser, but on 300 test papers a one-point change is within split noise, so I would next (1) repeat over several random splits, (2) give the gate the neighbour count as an input, initialise it open and allow values above 1, and (3) penalise constant gates or evaluate specifically on high-degree papers.

AI use: Claude (Anthropic) suggested candidate designs, implemented the code and plots, and helped draft this report; I chose the design and am responsible for the hypothesis and conclusions.